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Explainable ensemble learning model for cardiovascular disease prediction with feature optimization and data balancing

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Abstract

Cardiovascular disease (CVD) remains a leading global cause of death, highlighting the need for timely and accurate prediction methods. Traditional diagnostic approaches are often time-consuming, costly, and prone to human error, motivating the use of machine learning (ML) for improved risk assessment. However, ML-based CVD prediction faces challenges such as outliers, irrelevant features, class imbalance, and limited interpretability. In this study, we propose a comprehensive framework that addresses these challenges through advanced data preprocessing, feature selection, data balancing, ensemble learning, and explainable AI (XAI). Outliers are detected and corrected using the Z-score method, while XGBoost-based feature selection identifies the top 9 out of 13 predictors, enhancing model efficiency. Class imbalance is mitigated using techniques such as SMOTE, SMOTETomek, and SMOTE+RUS. Eight ML models - including Logistic Regression, KNN, SVM, Random Forest, XGBoost, Gradient Boosting, AdaBoost, and Gaussian Naive Bayes - are developed and rigorously evaluated using Accuracy, Precision, Recall, F1 score, and AUC-ROC metrics. A voting ensemble combining the top three models (Random Forest, Logistic Regression, and AdaBoost) achieves 98.63% accuracy and 99.13% AUC-ROC on the Cleveland Heart Disease dataset, demonstrating superior performance over individual models. SHAP and LIME analyses provide interpretability, allowing healthcare practitioners to understand model predictions and support clinical decision-making. This work contributes a robust, interpretable, and high-performing ML framework for early CVD detection, offering improved patient risk stratification and potential clinical integration.

Keywords Cardiovascular disease, Heart disease prediction, Cleveland dataset, Machine learning, Ensemble learning

1 Introduction

CVD remains a dominant public health disorder worldwide, with approximately 17.9 million deaths reported every year, accounting for nearly 31% of all deaths globally [1]. Despite advancements in the field of medicine and awareness, the incidence of CVD is

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on the rise because of a cluster of contributing factors such as urbanization, physical inactivity, unfavorable diet, and increasing population age around the globe [2]. Apart from having a high mortality rate, CVD also poses a significant healthcare burden, which is reflected in increased hospitalizations, long-term disabilities, and heavy economic burden. Therefore, timely and accurate diagnosis of CVD is of prime importance to reduce mortality and health expenditures and improve patient outcomes.

Previously, the diagnosis of CVD relied on physical check-ups, clinical expertise, patient history, laboratory tests, and imaging techniques such as echocardiograms and coronary angiograms. Although commonly used, these diagnostic methods have several limitations, such as high costs, lengthy evaluations, dependence on skilled medical professionals, and exposure to possible human error [3]. In addition, conventional diagnostic approaches often fail to provide the immediate and precise information needed for timely intervention, which is essential for preventing severe cardiovascular events. Consequently, scientists and clinicians are increasingly exploring AI and ML as novel tools to enhance CVD prediction and diagnosis.

The application of machine learning and deep learning methods for disease prediction has been emphasized by recent advancements in AI for healthcare. Studies concentrating on biological data and the prediction of cardiovascular disease have shown comparable advancements. Systematic reviews stress the importance of robust validation and generalizability [4], and optimized deep learning models have improved predictive performance [5]. According to [6], hyperparameter-tuned machine learning models have also demonstrated improved clinical applicability and predictive accuracy. The objectives of this study are in line with the importance of precise, comprehensible, and therapeutically applicable AI models, which are highlighted by these investigations taken together.

ML has been found to possess excellent abilities in handling large volumes of patient data, finding subtle trends, and making predictive models much better than standard statistical techniques [7]. Based on patient characteristics, such as demographic data, lifestyle, clinical data, and history, ML models can discover implicit correlations that are not easily evident to clinicians. This ability to learn and analyze large sets of data allows ML-based models to make more timely and accurate predictions, thereby facilitating the early detection and preventive measures against CVD.

However, despite its potential, the use of ML in cardiovascular medicine is marred by significant challenges. One of the serious problems is the selection of features since medical data are full of irrelevant and redundant features, which can lead to overfitting or underfitting and hence reduce the reliability of the predictive model [8]. The other serious problem is class imbalance, where data tend to have many more non-CVD cases than positive CVD cases. This imbalance causes biased models that fail to recognize high-risk patients, which translates to a high false-negative rate, which is particularly risky in a clinical setting [9]. Further, the uninterpretability of most ML models is a significant barrier to their deployment in a clinical setting. Most of the highest performing models, such as DL networks, are "black boxes," and therefore, it is difficult for physicians to interpret or trust their predictions [10].

As a step to reduce such issues, this study proposes a new approach of CVD prediction with the integration of feature selection, data balancing, and a voting ensemble machine learning model and explainable AI (XAI) methods. Feature selection protocols

allow the model's performance to be maximized by eliminating unnecessary variables and maximizing generalization. Data balancing strategies such as SMOTE and cost-sensitive learning allow predictive models to be ensured that do not over-sample the majority class and are able to accurately recognize CVD cases. In addition, ensemble learning protocols particularly voting-based ones - merge the individual strengths of a variety of classifiers to maximize robustness and predictability accuracy [11].

One of the standout aspects of this research is the integration of XAI techniques to improve the interpretability and transparency of machine learning predictions. Tools like SHAP and LIME offer detailed, model-agnostic insights into how predictions are made, helping to identify the most influential features in each case [12]. By incorporating these techniques, the study not only enhances model accountability but also seeks to bridge the gap between data-driven predictions and real-world clinical practice. This empowers healthcare professionals to better understand, trust, and act upon the AI-generated results, ultimately supporting more informed and reliable decision-making in cardiovascular disease diagnosis.

In a nutshell, this study focuses on developing a robust and interpretable ML-based CVD prediction model by addressing key issues in feature selection, data imbalance, and model interpretability. By applying ensemble learning and interpretable AI approaches, the present study seeks to attain greater prediction accuracies, solidify early detection research, and promote the clinical applicability of ML models to cardiovascular treatment. Through this, it is our desire to advance more reliable, interpretable, and clinically relevant AI-supported diagnostic models for CVD, thereby providing improved patient care and reducing the global prevalence of cardiovascular diseases.

1.1 Technical challenges

Although the healthcare issue is crucial, it is just as crucial to solve the technical issues around ML models for the prediction of CVD. These challenges are as follows.

- High dimensionality of medical data makes it challenging to identify the most relevant features for cardiovascular disease prediction.
- Class imbalance in the dataset, where healthy individuals significantly outnumber diseased individuals, can lead to biased models and reduced prediction accuracy.
- Lack of transparency and interpretability in machine learning models, hindering their adoption in real-world clinical settings.

This study aims to address these challenges by integrating feature selection, data balancing, voting ensemble methods, and explainable AI (XAI) to build a robust and interpretable CVD prediction model.

1.2 Problem statement

Predicting CVD using ML is hindered by the challenges of selecting the most relevant features, dealing with imbalanced datasets, and ensuring that the model's predictions are interpretable. Standard machine learning models can struggle with overfitting or underfitting due to irrelevant or redundant features. Furthermore, when data classes are imbalanced, models may fail to detect the minority class (disease occurrence), leading to a high false-negative rate. Moreover, many machine learning models lack transparency, making it difficult for healthcare professionals to trust or understand the results.

1.3 Research objectives

The general objective of this study is to develop a robust, accurate, and interpretable machine learning (ML) model for predicting cardiovascular disease (CVD) by addressing key technical challenges. Each research objective is directly aligned with a corresponding challenge as outlined below:

- **Challenge: High-dimensional and noisy features Objective: Feature Selection** – Identify the most significant features that contribute to CVD prediction, thereby reducing dimensionality while retaining predictive power.
- **Challenge: Severe class imbalance in CVD datasets Objective: Data Balancing** – Apply balancing techniques (e.g., SMOTE, undersampling, hybrid approaches) to improve the model's ability to correctly classify both majority and minority classes.
- **Challenge: Limited accuracy and generalization of single models Objective: Ensemble Modeling** – Build a voting ensemble by combining multiple learners to enhance predictive accuracy and generalization.
- **Challenge: Lack of model transparency in healthcare AI Objective: Explainable AI (XAI)** – Incorporate explainability methods such as SHAP and LIME to provide transparent and actionable insights for healthcare professionals.
- **Challenge: Need for reliable validation of predictive performance Objective: Model Evaluation** – Evaluate model performance using multiple metrics (accuracy, precision, recall, F1-score, AUC-ROC, MSE, MAE, RMSE, MCC, and Kappa), with particular emphasis on minority class outcomes.

1.4 Contributions

The following are the key contributions of this study:

- Comprehensive Feature Selection Approach: Propose a systematic feature selection methodology to enhance predictive accuracy while minimizing computational complexity by identifying key variables in the Cleveland dataset.
- Effective Data Balancing Strategies: Investigate and implement effective data balancing methods to address the challenges of class imbalance, ensuring the model provides accurate predictions across both classes.
- Voting Ensemble Model for Enhanced Performance: Design and implement a voting ensemble model that leverages the strengths of multiple base learners to create a more robust and accurate prediction system for cardiovascular disease.
- Integration of Explainable AI: Incorporate modern explainability tools such as SHAP or LIME to offer transparent and understandable predictions, fostering trust in the machine learning model from medical professionals.

1.5 Research questions

In this study, we aim to tackle key challenges in cardiovascular disease prediction specifically, feature selection, class imbalance, and model interpretability through the application of ML techniques. To guide this investigation, we focus on the following research questions:

1. What are the most relevant features for predicting cardiovascular disease in the Cleveland dataset?

2. How can class imbalance in the dataset be effectively addressed to improve prediction accuracy?
3. Which data balancing technique (oversampling, undersampling, or synthetic data generation) yields the best results for CVD prediction?
4. How does the performance of a voting ensemble model compare to individual machine learning models in terms of prediction accuracy and generalization?

The remainder of this paper is structured as follows: Sect. 2 provides a concise review of related work, positioning this study within the broader scope of existing research. Section 3 outlines the proposed methodology and details the datasets utilized. Section 4 describes the experimental framework and offers a comprehensive evaluation of the machine learning models implemented. Finally, Sect. 5 presents the key conclusions drawn from the study and highlights its implications for future research in cardiovascular disease prediction.

2 Related works

The Cleveland Heart Disease dataset from the UCI Machine Learning Repository has been extensively used in CVD prediction research. To enhance predictive accuracy and address the limitations of traditional approaches, researchers have explored a wide array of ML and data-driven techniques. This section provides an overview and critical analysis of prior studies related to heart disease detection using machine learning models, highlighting their strengths, limitations, and contributions. A variety of algorithms have been employed as essential tools for predictive modeling across medical domains, with the core objective of building models that can effectively predict whether an individual is likely to develop cardiovascular disease. Significantly, many past studies have focused on the Cleveland dataset for training and evaluating these models. The current research builds upon the foundation laid by earlier works, drawing from their insights to further refine and advance the field of CVD prediction.

Korial et al. [13] proposed an enhanced ensemble-based approach for detecting CVD by integrating Chi-Square feature selection with multiple ML models. The study was conducted using the Cleveland heart disease dataset, which includes 303 instances and 13 features. Initially, data normalization was applied, followed by the Chi-Square algorithm to identify the top five most significant features contributing to heart disease prediction. Four individual ML models LR, RF, NB, and K-KNN were trained separately. Their predictions were then combined using a majority voting ensemble strategy. The results revealed that the ensemble model achieved an accuracy of 86.84% before feature selection, which significantly improved to 92.11% after applying the Chi-Square method. This performance also marked a 3.84% improvement over the previously best-performing model in the literature, highlighting the benefit of integrating feature selection with ensemble learning for effective CVD prediction.

Kumar et al. [14] carried out a comparative analysis to assess the effectiveness of several ML algorithms namely KNN, SVM, ANN, and CNN in predicting CVD using the UCI Cleveland heart disease dataset. Their approach included data exploration, feature engineering, and model training, with performance evaluated through accuracy, precision, and recall metrics. Among all the models tested, the CNN-based model delivered the best performance, achieving 83.61% accuracy, 76.0% precision, and a notably high 97.0% recall. In comparison, KNN yielded 66.7% accuracy, 88.8% precision, and 91.7%

recall; SVM with a linear kernel achieved 74.2% accuracy, 74.0% precision, and 85.0% recall; while ANN reached 70.08% accuracy, 77.0% precision, and 84.2% recall. These results suggest that deep learning models like CNN can offer significant improvements over conventional machine learning techniques in heart disease prediction tasks.

Chandrasekhar et al. [15] developed a soft voting ensemble classifier aimed at enhancing the prediction of CVD using data from both the Cleveland and IEEE Dataport heart disease datasets. Their methodology began with comprehensive data preprocessing, including the removal of missing and duplicate entries, followed by feature scaling to ensure consistent input across models. They employed six ML algorithms RF, KNN, LR, NB, GB, and ADB. and fine-tuned them using GridSearchCV with 5-fold cross-validation to achieve optimal configurations. The models were assessed using key performance metrics such as accuracy, precision, recall, and F1-score. While LR and ADB individually performed best among the single models, the soft voting ensemble, which integrated predictions from all six classifiers, delivered the highest overall accuracy 93.44% on the Cleveland dataset and 95% on the IEEE Dataport dataset. These findings underscore the strength of ensemble methods, particularly soft voting, in improving diagnostic accuracy for heart disease prediction.

Abdellatif et al. [16] proposed an effective ML framework for detecting CVD and classifying its severity level using the Cleveland dataset. To address class imbalance, the authors employed the SMOTE. The study evaluated six ML classifiers, including an ET classifier optimized using Hyperband, alongside comprehensive hyperparameter tuning. Two public datasets were used for model development and testing. The optimized ET classifier, when combined with SMOTE, achieved outstanding performance attaining 99.2% and 98.52% accuracy in disease detection across the datasets. Furthermore, the model achieved 95.73% accuracy in classifying the severity of CVD using the Cleveland dataset, surpassing previous state-of-the-art techniques in both detection and classification tasks.

Tama et al. [17] introduced an innovative approach for detecting coronary heart disease (CHD) using a two-tier ensemble learning framework designed to boost predictive performance. In their method, powerful ensemble models RF, GB Machine, and XGBoost were combined within a stacked architecture, acting as base learners. To ensure the model's reliability and generalizability, it was tested across four widely recognized datasets: Hungarian, Cleveland, Statlog, and Z-Alizadeh Sani,. The most influential features from each dataset were selected using PSO, helping to reduce dimensionality and focus on the most impactful variables. To validate the statistical significance of their results, a two-fold hypothesis test was performed. Using 10-fold cross-validation, the proposed model consistently outperformed both single classifiers and traditional ensemble techniques in terms of accuracy, F1-score, and AUC. These results demonstrate the model's robustness and its potential to significantly enhance CHD prediction in clinical applications.

Rahim et al. [18] introduced an integrated framework called MaLCaDD, which combines LR and KNN to enhance the prediction of CVD. Their approach was tested on the Cleveland dataset and demonstrated impressive performance, achieving an accuracy of 95.5%. The study incorporated SMOTE to handle class imbalance and utilized feature selection techniques to retain only the most informative attributes. The results suggest

that MaLCaDD offers a reliable and efficient solution for early diagnosis of heart disease, highlighting its potential in clinical decision support systems.

Noroozi et al. [19] conducted a detailed study on the impact of various feature selection techniques on the performance of ML models for CVD prediction. Using the Cleveland Heart Disease dataset, they applied sixteen feature selection methods across three main categories: filter-based, wrapper-based, and evolutionary approaches. They evaluated the predictive performance of seven ML algorithms Bayes Net, Naïve Bayes, Multivariate Linear Model, SVM, Logit Boost, J48, and Random Forest using several metrics such as ROC Area, Precision-Recall Curve (PRC), Specificity, Accuracy, Sensitivity, F-measure, and Precision. Their findings highlighted that the choice of feature selection method significantly influences the accuracy of the models, with some combinations outperforming others. Among the tested setups, SVM combined with filter-based feature selection achieved the highest accuracy of 85.5%, showing the importance of tailored feature engineering in predictive modeling.

Ahamad et al. (2023) [20] explored heart disease prediction using both the UCI Cleveland dataset and a merged dataset that included records from Cleveland, Hungary, Switzerland, and Long Beach V. They evaluated the performance of six different ML algorithms: LR, KNN, SVM, DT, RF, and XGBoost. Among these models, the SVM classifier, when fine-tuned with optimized hyperparameters, achieved the highest test accuracy of 87.91% on the UCI Cleveland dataset. This demonstrates the effectiveness of SVM in handling heart disease prediction tasks when appropriately tuned.

Ogundepo et al. (2022) [21] conducted a performance comparison study using two publicly available heart disease datasets: the Cleveland dataset for training and the Statlog dataset for validation. A detailed exploratory data analysis was performed on the Cleveland dataset, which included a Chi-square test of independence to evaluate feature relationships. Ten widely used classification algorithms were then applied to predict heart disease classes. To ensure robustness, the dataset was randomly split 200 times, allocating 70% (208 samples) for training and 30% (89 samples) for testing in each iteration. Among all models, the SVM emerged as the most effective, achieving 85% accuracy, 82% sensitivity, 88% specificity, 87% precision, an AUC of 91%, and a log loss of 38%. These results highlight SVM's strong potential for heart disease prediction.

Shrestha et al. [22] investigated the prediction of CVD using a range of ML techniques, including RF, GB, XGBoost, LR, and LSTM networks. Prior to model training, the dataset was thoroughly preprocessed this involved addressing missing values, encoding categorical features, and converting the target variable into a binary format. The performance of the models was assessed using key metrics such as accuracy, precision, recall, F1-score, and AUC-ROC. To enhance model transparency, SHAP values were employed to explain the impact of individual features on predictions. Among all the models tested, XGBoost demonstrated the highest performance, achieving an AUC-ROC score of 0.94 and an accuracy of 90%, indicating its strong potential in cardiac disease prediction tasks.

In order to increase the accuracy of heart disease detection, Nursyahrina et al. (2024) [23] designed an RNN intelligent system that combines neural networks with rough set theory. Nine important features were chosen for model training from the Cleveland Heart Disease Dataset using the rough set component, which helped to simplify

the data. The RNN model performed better than conventional neural networks in comparison evaluations, with an accuracy of 88.52%, an F1-score of 88.14%, and an AUC of 88.85%.

Nazari et al. [24] proposed a novel ensemble learning model for CVD prediction and key risk factor identification, utilizing both the Cleveland and Noor datasets. Their approach involved combining multiple classifiers, with the ensemble weights optimized using a genetic algorithm. The proposed model achieved precision scores of 88.05% on the Cleveland dataset and 90.12% on the Noor dataset, demonstrating its effectiveness in enhancing early detection and providing deeper insights into the contributing factors of CVD.

With an emphasis on heart failure prediction, Reddy et al. [25] used ML approaches, such as DT and GNB, on the UCI Cleveland dataset. Examining how well these models work to use clinical risk indicators to aid in medical decision-making was the main goal. With an accuracy of 86.0% on the Heart Failure Dataset, the GNB algorithm outperformed the other models that were assessed. According to the results, GNB is an effective and appropriate method for the early identification of cardiac disorders.

Most existing studies on cardiovascular disease prediction focus on boosting accuracy but often overlook several important challenges. Many do not carefully select the most relevant features, which can leave noise in the data, or fail to address class imbalance, which makes minority cases harder to detect. Likewise, single-model approaches are common, but they may not generalize well across different datasets. Another major limitation is the lack of transparency, few studies use explainable AI methods to make their predictions understandable for clinicians. Finally, many works rely on limited evaluation metrics, giving an incomplete picture of model performance.

To close these gaps, this study introduces a complete pipeline that combines data preprocessing, feature selection, balancing techniques, ensemble learning, and explainable AI. This design aims not just for higher accuracy but also for a model that is robust, fair across classes, and interpretable, making it more practical for real-world clinical decision-making.

3 Methodology

As part of this study's methodology, ML algorithms have been utilized to analyze clinical data and forecast CVD. A methodical procedure is summed up in the block diagram shown in Fig. 1. The suggested approach involves a pipeline systematically enhancing the accuracy, stability, and interpretability of the predictions related to CVD. Firstly, the removal of outliers is conducted via the Z-score method to purify the dataset and enhance the reliability of the model. Then, feature selection is conducted through the XGBoost (XGB) feature importance method to keep only the most predictive features to guarantee less noise and computational cost. To handle class imbalance, common in medical datasets, three resampling techniques - SMOTE, SMOTETomek, and SMOTE+RUS are employed to enable the models to learn effectively from both the majority and minority classes. Eight classical machine learning models (LR, KNN, SVM, RF, XGB, GB, ADB, and GNB) are trained and evaluated on several performance metrics. An ensemble voting model is then created from the three best-performing classifiers, thus taking advantage of their strengths together to improve predictive capability. Finally, to reduce the opaque nature intrinsic to ensemble models, Explainable Artificial

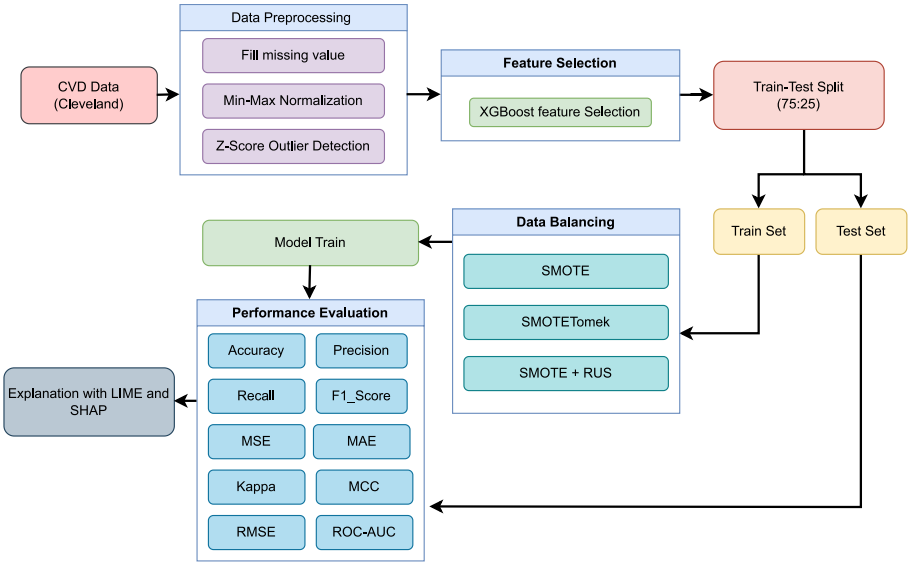


Fig. 1 Brief Overview of the proposed methodology

Intelligence techniques LIME and SHAP are integrated, allowing transparency for the model's decisions and fostering clinical interpretability. This combined approach offers an extremely accurate, interpretable, and clinically relevant system for the detection and risk prediction of early CVD.

The main steps were:

1. **Data Collection**: A relevant heart disease dataset was sourced from publicly available online repositories, such as the UCI Machine Learning Repository, which is widely recognized for hosting benchmark datasets in machine learning research.
2. **Data Preprocessing**: The data set was thoroughly inspected for completeness, consistency and the presence of outliers. Any missing or erroneous values were appropriately handled and the features were subsequently encoded and normalized to ensure suitability for machine learning algorithms.
3. **Feature Engineering**: The most significant features have been identified through feature selection processes.
4. **Train-Test split**: The dataset was divided into subsets for testing and training in order to make model construction and performance assessment easier.
5. **Data Balancing**: Balance the train data so that the ML models perform very well.
6. **Model Building**: To create predictive models, supervised ML techniques were used on the preprocessed data set.
7. **Model Evaluation**: The models were evaluated using a comprehensive set of performance metrics.
8. **Model Comparison and best model selection**: The prediction capacity of the top-performing models was next compared with that of recent, state-of-the-art studies. Based on comprehensive evaluation metrics, the best model for predicting CVD was selected.
9. **Model Explanation**: The Selected model has been explained with LIME and SHAP.

The methodology outlined above serves as the study's methodological framework. Additional details regarding the dataset, its preprocessing, and the evaluation criteria employed are given in the following sections.

3.1 Dataset

For this study, the Cleveland Heart Disease dataset, a widely recognized benchmark in CVD research, has been employed. The Cleveland dataset is widely used in CVD research due to its well-established status as a benchmark dataset. It offers a rich collection of features that cover various aspects of heart health, making it suitable for complex predictive modeling. The dataset was sourced from [26] and consists of 303 patient records, each comprising 14 clinically relevant features. These attributes cover a broad spectrum of health indicators, including demographic information (such as age and sex), medical measurements (like cholesterol levels and blood pressure), and clinical symptoms (such as chest pain type). A well defined goal variable that indicates the existence or absence of CVD is also included in the dataset. A detailed overview of the dataset's attributes is presented in Table 1.

3.2 Data cleaning and preparation

The initial step after acquiring the datasets is to preprocess them before incorporating them into the machine learning models. Data preparation is an essential stage since real-world data is frequently inconsistent, lacking in specific behaviors, and full of errors and outliers. All of these issues can be resolved, and the dataset can be prepared for usage in any machine-learning model through data preprocessing. A machine learning model will not be able to produce a better result when it is trained on an unprocessed dataset, and its performance will eventually decline noticeably. The preparation of data is a method that requires a number of stages to be completed, such as altering the data format. The following are the data preprocessing steps used in this study:

1. Identifying and removing duplicate records.
2. Features containing missing values were subsequently identified and filled with a median filter.

Table 1 Description of features in the cleveland heart disease dataset

Feature	Description
`Age`	Age of the patient in years
`Sex`	Sex of the patient (1 = male, 0 = female)
`cp`	Chest pain type (0 = typical angina, 1 = atypical angina, 2 = non-anginal pain, 3 = asymptomatic)
`Trestbps`	Resting blood pressure (in mm Hg)
`chol`	Serum cholesterol level (in mg/dL)
`fbs`	Fasting blood sugar> 120 mg/dL (1 = true, 0 = false)
`restecg`	Resting electrocardiographic results (0 = normal, 1 = ST-T abnormality, 2 = left ventricular hypertrophy)
`thalach`	Maximum heart rate achieved
`exang`	Exercise-induced angina (1 = yes, 0 = no)
`oldpeak`	ST depression induced by exercise relative to rest
`Slope`	Slope of the peak exercise ST segment (0 = upsloping, 1 = flat, 2 = downsloping)
`ca`	Number of major vessels (0–3) colored by fluoroscopy
`thal`	Thalassemia (1 = normal, 2 = fixed defect, 3 = reversible defect)
`Target`	Presence of heart disease (0 = no, 1 = yes)

Table 2 Class distribution before and after Z-Score normalization

Method	Class Label	
	0	**1**
Before Z-Score	164	139
After Z-Score	162	132

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63	1	0	145	233	1	2	150	0	2.3	2	0	2	0
1	67	1	3	160	286	0	2	108	1	1.5	1	3	1	1
2	67	1	3	120	229	0	2	129	1	2.6	1	2	3	1
3	37	1	2	130	250	0	0	187	0	3.5	2	0	1	0
4	41	0	1	130	204	0	2	172	0	1.4	0	0	1	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
298	45	1	0	110	264	0	0	132	0	1.2	1	0	3	1
299	68	1	3	144	193	1	0	141	0	3.4	1	2	3	1
300	57	1	3	130	131	0	0	115	1	1.2	1	1	3	1
301	57	0	1	130	236	0	2	174	0	0.0	1	1	1	1
302	38	1	2	138	175	0	0	173	0	0.0	0	0	1	0

303 rows × 14 columns

Fig. 2 Initial data frame of the cleveland dataset

3. The Z-Score method with a threshold value = 3 was used to find out the outlier, and subsequently, remove those with outliers. A statistical metric called the Z-score counts the number of standard deviations that separate a data point from the dataset mean. The following formula is used to compute it:

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

where x represents the data point, μ is the mean of the dataset, and σ is the standard deviation. After the outlier detection, it is found that there were 9 records with outliers; the dataset no longer included these records. Table 2 shows the class distribution before and after applying Z-Score normalization. Initially, there were 164 samples in class 0 and 139 samples in class 1. After normalization, the counts decreased to 162 samples in class 0 and 132 samples in class 1. This indicates that a total of 9 records were removed during the normalization process - specifically, 2 from class 0 and 7 from class 1. Such removal likely resulted from filtering out samples with invalid or extreme values during standardization.

4. The Dataset has been normalized using min-max normalization techniques so that the range of data in the dataset is in a shorter range. it will help the ML model to train well.

The Cleveland dataset's original and preprocessed data frames are shown in Figs. 2 and 3, respectively. This graphic depiction shows that there are 303 rows, or instances, and 14 attributes, or columns, in the dataset. Curiously, since all attribute values are already in a numerical format, conversion is not required.

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	0.708333	1	0.000000	0.481132	0.244292	1	1.0	0.603053	0	0.370968	1.0	0.000000	0.5	0
1	0.791667	1	1.000000	0.622642	0.365297	0	1.0	0.282443	1	0.241935	0.5	1.000000	0.0	1
2	0.791667	1	1.000000	0.245283	0.235160	0	1.0	0.442748	1	0.419355	0.5	0.666667	1.0	1
3	0.166667	1	0.666667	0.339623	0.283105	0	0.0	0.885496	0	0.564516	1.0	0.000000	0.0	0
4	0.250000	0	0.333333	0.339623	0.178082	0	1.0	0.770992	0	0.225806	0.0	0.000000	0.0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
298	0.333333	1	0.000000	0.150943	0.315068	0	0.0	0.465649	0	0.193548	0.5	0.000000	1.0	1
299	0.812500	1	1.000000	0.471698	0.152968	1	0.0	0.534351	0	0.548387	0.5	0.666667	1.0	1
300	0.583333	1	1.000000	0.339623	0.011416	0	0.0	0.335878	1	0.193548	0.5	0.333333	1.0	1
301	0.583333	0	0.333333	0.339623	0.251142	0	1.0	0.786260	0	0.000000	0.5	0.333333	0.0	1
302	0.187500	1	0.666667	0.415094	0.111872	0	0.0	0.778626	0	0.000000	0.0	0.000000	0.0	0

294 rows × 14 columns

Fig. 3 Preprocessed data frame of the cleveland dataset

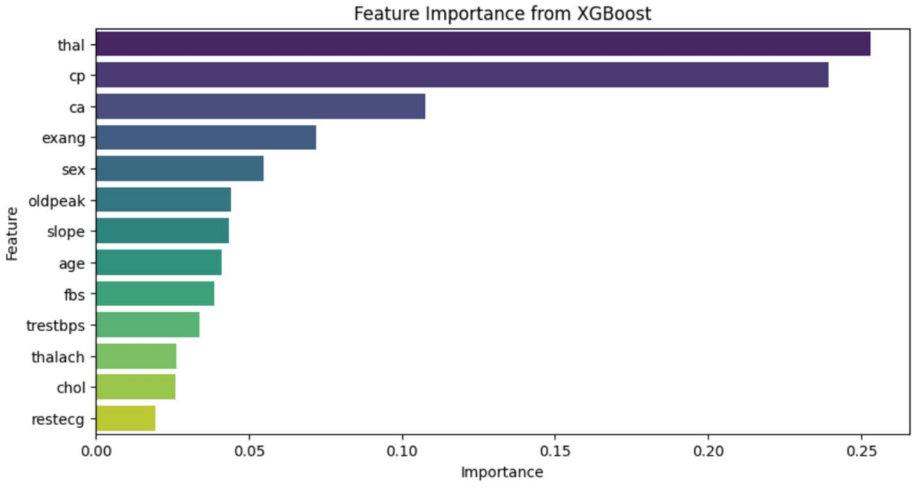


Fig. 4 Feature Importance of the dataset

3.3 Feature selection

In machine learning, feature selection is required for enhancing model effectiveness by the removal of redundant or unnecessary features, which may result in overfitting and very complex models. The use of feature selection minimizes both computational time and cost associated with training, thereby making models more efficient. By concentrating on the most significant features, feature selection helps improve the interpretability of models as well as improved generalization to new, unseen data. Also, it can render the model more precise and less susceptible to noise, harming its predictions. In this study, the XGBoost feature selection technique was used to select the key features for the train set of data. Figure 4 shows the feature importance of the dataset. The XGBoost feature importance plot identifies the top determinants of the prediction of the target variable, and thal, cp, and ca are the top contributors. Thalassemia (thal) and chest pain type (cp) stand out in particular, perhaps because they both have significant correlations with heart disease, while ca (number of main vessels colored by fluoroscopy) is at the heart of the diagnosis. Moderately important features, including exang (exercise-induced angina), sex, oldpeak (ST depression), and slope (ST segment slope), are very important, indicating their significance in cardiovascular well-being. Less important features include age, fbs (fasting blood sugar), trestbps (resting blood pressure), thalach

Table 3 Features and their importance

Feature	Importance					
	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
thal	0.3431	0.2839	0.3331	0.2272	0.2801	0.2935
cp	0.1192	0.1896	0.1457	0.1600	0.1619	0.1553
ca	0.1088	0.1239	0.1215	0.1310	0.1097	0.1190
Exang	0.0750	0.1044	0.0602	0.0896	0.0715	0.0802
Sex	0.0604	0.0657	0.0582	0.0595	0.0640	0.0616
oldpeak	0.0541	0.0480	0.0563	0.0591	0.0619	0.0559
slope	0.0485	0.0434	0.0456	0.0515	0.0507	0.0480
Age	0.0476	0.0388	0.0417	0.0473	0.0403	0.0431
fbs	0.0467	0.0325	0.0338	0.0443	0.0385	0.0392
trestbps	0.0300	0.0250	0.0319	0.0434	0.0368	0.0334
thalach	0.0281	0.0202	0.0296	0.0295	0.0313	0.0277
chol	0.0253	0.0156	0.0279	0.0291	0.0270	0.0250
restecg	0.0130	0.0091	0.0143	0.0285	0.0264	0.0183

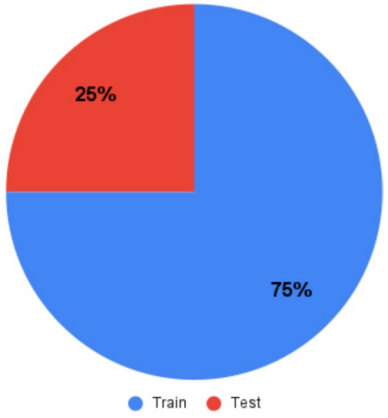


Fig. 5 Train-Test split of the dataset

(maximum heart rate), chol (cholesterol), and restecg (resting ECG results), indicating that they have less predictive capability or are dominated by stronger features. XGBoost plots feature importance based on the frequency of a feature being used for splitting and its contribution to the reduction of model error, and thus, this selection is paramount in identifying major predictors in heart disease classification [27].

Based on the feature importance analysis in table 3, the top nine features selected for cardiovascular disease prediction are *thal, cp, ca, exang, sex, oldpeak, slope, age,* and *fbs*. The average importance of the ninth-ranked feature (*fbs*) is 0.0392, which is considered as the threshold; thus, all features with an average importance greater than or equal to 0.039 were selected for the final model.

3.4 Train-test split

Two subsets of the dataset were randomly selected: 25% for testing and 75% for training. As a result, 221 samples were utilized to train the model, and 73 examples were set aside to assess the model's performance. The training set was used to build all machine learning models, while the testing set was used to evaluate the models' capacity for generalization and prediction accuracy. The dataset split is visually represented in Fig. 5.

3.5 Data balancing

The original dataset is imbalanced. ML models work well with a balanced dataset. We used a variety of data-balancing strategies to resolve the class imbalance in our datasets. By improving the distribution of the majority and minority classes, these approaches guarantee that the models are trained on data that is balanced. Specifically, we tried out the following techniques:

SMOTE: SMOTE is a powerful technique for imbalanced classification problems that can be used to balance the data by generating synthetic instances for the minority class. It's extremely useful when dealing with datasets where one class has much fewer instances than the other [28]. The process of creating a synthetic sample is described as:

$$X_{\text{new}} = X_s + \delta \cdot (X_k - X_s) \tag{2}$$

where:

- X_{new} represents the newly generated synthetic instance.
- X_s is an example from the minority class in the dataset..
- X_k is a randomly chosen neighbor of X_s from the same minority class.
- δ s a randomly selected value between 0 and 1, used to introduce variation in the synthetic data creation.**SMOTETomek:** SMOTETomek is a hybrid resampling technique designed to address class imbalance while also improving data quality. It combines two powerful methods: SMOTE, which creates synthetic examples for the underrepresented class, and Tomek Links, which identifies and removes borderline or overlapping samples that may cause confusion during classification. This combination not only balances the dataset but also helps clean it by eliminating noise and ambiguity. SMOTETomek is especially effective in imbalanced classification problems like those often encountered in medical datasets where achieving both balance and clarity is essential for reliable model performance.

SMOTE + RUS: SMOTE + RUS is a hybrid resampling approach that combines SMOTE with RUS to more effectively handle class imbalance. By generating synthetic examples of the minority class and simultaneously reducing the majority class, this method strikes a balance between retaining useful data and addressing the skewed class distribution–leading to better model performance, especially in imbalanced datasets like those often found in medical diagnosis. SMOTE generates synthetic minority class instances, and random Under-Sampling (RUS) is employed to reduce the majority class instances. This helps achieve a good balance of the dataset without the pitfalls of implementing either approach separately, such as overfitting from uncontrolled oversampling or data loss from excessive undersampling. RUS decreases the number of samples in the majority class by randomly eliminating samples, and, thereby, inhibits model bias towards the majority class [29]. Following the application of both techniques, the balanced dataset thus obtained can be expressed as:

$$D' = D_1 \cup D_1^{\text{SMOTE}} \cup D_0' \tag{3}$$

where:

- D_1: original minority class samples.
- D_1^{SMOTE}: synthetic minority class samples.

- $D_0' \subset D_0$: reduced majority class samples selected by RUS.

In this study, SMOTE, SMOTETomek, and SMOTE+RUS were employed to address class imbalance while preserving the overall quality of the dataset. SMOTE generated synthetic samples for the minority class to mitigate overfitting, SMOTETomek refined the dataset by removing noisy and overlapping points to achieve clearer class boundaries, and SMOTE+RUS combined oversampling with undersampling to prevent dominance of the majority class. Unlike simple random sampling techniques, which can result in overfitting or data loss, or more complex methods such as ADASYN, which may introduce additional noise, these approaches provided a practical and widely accepted trade-off between performance and reliability. We chose SMOTE oversampling over class weighting because it generates synthetic minority samples instead of duplicating existing ones, reducing the risk of overfitting. Random oversampling or heavy class weighting can cause models to memorize repeated instances, whereas SMOTE creates new, plausible data points, improving generalization. This makes it especially effective for imbalanced datasets, providing diverse examples for the model to learn meaningful patterns. Table 4 indicates the number of samples in the train set for each data balancing technique.

3.6 Model selection

To identify the most effective algorithm for CVD prediction, eight widely recognized ML models were selected based on their strong track record in both medical and classification tasks. These include LR, KNN, SVM, RF, XGB, GB, ADB, and GNB. Each model brings unique advantages such as ease of interpretation, resilience to noisy data, or the ability to capture complex non linear patterns which makes them well-suited for a comprehensive comparison.

All models were trained and tested on the same preprocessed and balanced dataset to ensure fairness in evaluation. Their performance was assessed using a variety of metrics, including accuracy, precision, recall, F1-score, MSE, MAE, RMSE, MCC, ROC-AUC, and Kappa. Based on the evaluation results, the top performing three models were selected to form a voting ensemble, aimed at combining their strengths to improve overall generalization and reliability. Figure 6 provides a visual overview of the ML models used in this research.

3.6.1 Logistic regression (LR)

The LR model is a statistical and ML approach to binary classification problems. In actuality, it is a classification method rather than a regression methodology, despite its name. The likelihood that an instance belongs to a particular class is predicted by LR. The logistic (sigmoid) function is used to achieve this:

Table 4 Number of samples in Train dataset on different data balance techniques

Method	Class Label	
	0	1
NONE	122	99
SMOTE	122	122
SMOTETomek	115	115
SMOTE + RUS	109	109

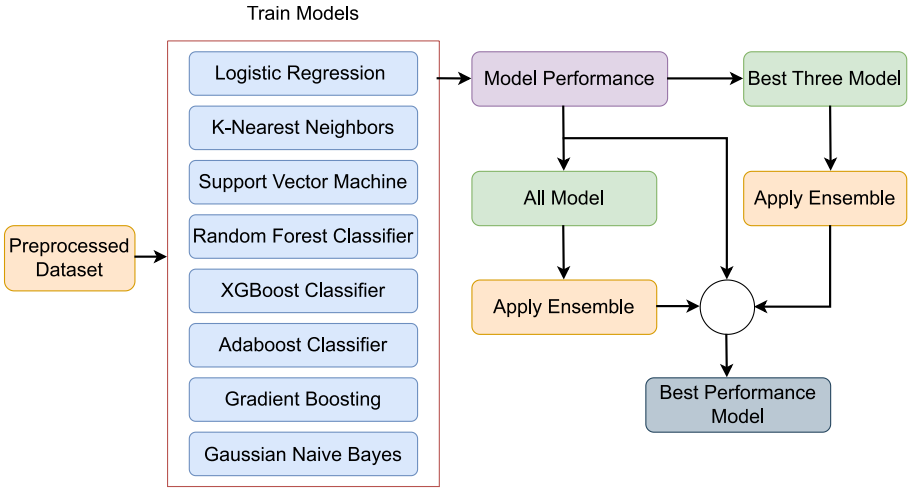


Fig. 6 Trained ML models and model selection procedure

$$\sigma(z) = \frac{1}{(1 + e^{-z})} \tag{4}$$

Where: $z = w_0 + w_1x_1 + w_2x_2 + \ldots\ldots\ldots + w_nx_n$ ((Linear combination of input features and weights) and a value between 0 and 1 is produced by $\sigma(z)$, and this value is regarded as the probability of the positive class.

3.6.2 K-nearest neighbors (KNN)

A popular supervised machine learning algorithm for classification and regression applications, KNN is straightforward but efficient. Based on the average value (for regression) or majority class (for classification) of the k closest data points in the feature space, it generates predictions. Using a lazy learning approach and non-parametric characteristics, the algorithm predicts based on the relative similarity of input points. KNN uses the majority vote of the k nearest neighbors to determine a class label in classification, while it uses the mean value of the nearby data points to predict the output in regression.

3.6.3 Support vector machine (SVM)

A popular supervised learning technique for classification, regression, and outlier identification applications is SVM. It is especially good at handling both linear and non-linear classification issues because it uses kernel functions that project data into higher-dimensional regions, making class separation easier. SVM is known for its robustness, especially when working with high-dimensional datasets, which makes it well-suited for complex applications like medical diagnosis and disease prediction [30]..

3.6.4 Random forest (RF)

The RF, an ensemble learning method, constructs many decision trees during training. For classification problems, it aggregates all of the trees' predictions and selects the majority vote as the final output, whereas for regression tasks, it takes the average of all the trees' forecasts. This approach improves accuracy, reduces overfitting, and increases model robustness by leveraging the diversity of the individual trees. Averaging the outputs of many trees increases the predictive power and decreases the risk of overfitting. This makes it a strong contender in regression and classification.

3.6.5 XGBoost

A strong and effective gradient boosting algorithm implementation that prioritizes speed, scalability, and performance is Extreme Gradient Boosting (XGBoost). It stands out for its advanced regularization techniques, which help reduce overfitting, and its ability to handle enormous datasets with high predictive accuracy. These qualities make XGBoost a popular choice in many ML tasks, including medical diagnosis and CVD prediction. It uses advanced features such as tree pruning, parallel computation, and handling of missing values to improve accuracy and efficiency. XGBoost has become a popular choice in healthcare applications, including CVD, due to its robustness and ability to handle structured tabular data effectively [31].

3.6.6 Gradient boosting

An effective ensemble learning technique called gradient boosting sequentially combines multiple weak learners, usually decision trees, to create a strong prediction model. By steadily enhancing the overall performance, each new model is trained to fix the errors created by the ones that came before it. For both classification and regression tasks, the approach is quite effective because it minimizes a selected loss function using gradient descent. With built-in mechanisms for regularization, Gradient Boosting not only enhances accuracy but also helps prevent overfitting. Thanks to these strengths, it has become a popular choice in healthcare applications, including the prediction of cardiovascular disease [32].

3.6.7 AdaBoost

Adaptive Boosting (AdaBoost) is a popular ensemble learning method that builds a strong classifier by combining several weak learners, usually simple models like decision stumps. The key idea behind AdaBoost is to give more attention to the training samples that are hard to classify. During each iteration, it adjusts the weights of the samples - boosting the importance of those that were misclassified - so that the next learner focuses more on those difficult cases. Over time, this process helps the ensemble correct its mistakes and improve overall accuracy, making AdaBoost a powerful tool for classification tasks. AdaBoost improves prediction accuracy by minimizing the overall classification error through weighted majority voting. Its robustness and ability to reduce bias make it a suitable algorithm for healthcare applications, including cardiovascular disease prediction [33].

3.6.8 Gaussian Naive Bayes

Gaussian Naive Bayes (GNB) is a simple yet effective probabilistic classifier that operates under the assumption that features are independent of one another. It models each feature using a Gaussian (normal) distribution and uses Bayes' Theorem to estimate the probability of each class based on the given input. Thanks to its lightweight nature and fast training time, GNB is especially suitable for small to medium-sized datasets. In biomedical contexts, including cardiovascular disease prediction, it has been widely used due to its efficiency and surprisingly good performance even with limited or noisy data [34]. Its transparency and ease of interpretation also make it a strong baseline model for healthcare-related classification tasks.

3.6.9 Ensemble omdel

Ensemble models are powerful machine learning approaches that combine the outputs of multiple individual learners to produce more accurate and reliable predictions. Rather than relying on a single model, ensemble methods leverage the complementary strengths of different algorithms to improve accuracy, reduce overfitting, and enhance model stability. Several widely used ensemble strategies exist. Bagging, as implemented in Random Forest, trains multiple models on random subsets of the data to minimize variance. Boosting techniques, such as AdaBoost and Gradient Boosting, iteratively focus on correcting the errors of previous models to reduce bias and improve performance. Voting methods aggregate the predictions of multiple models, either through majority voting or by averaging, to reach a consensus decision.

Ensemble models outperform individual models because they combine multiple learners to reduce both bias and variance, addressing the limitations of single models. Bagging methods (e.g., Random Forest) lower variance by averaging diverse predictors, while boosting methods (e.g., XGBoost) reduce bias by focusing on difficult cases. Empirical studies consistently show that ensembles achieve higher accuracy, robustness, and generalization, making them more reliable than individual models, especially in complex or imbalanced datasets.

In this research, a voting ensemble approach was adopted by combining the three best-performing classifiers. This strategy takes advantage of their individual strengths, resulting in a more reliable and accurate system for predicting cardiovascular disease [35].

- Voting: A machine learning model known as a voting classifier learns from a collection of models and forecasts an output according to the probability that the result will belong to the desired class. Figure 7 shows the basic procedure of the voting ensemble.

3.6.10 Models and their hyperparameters

The table 5 lists the different machine learning models used in the study along with the main hyperparameters chosen for each one. These hyperparameters control how each model learns from data. For example, Logistic Regression is set to use L2 regularization with the 'lbfgs' solver and a maximum of 100 iterations. K-Nearest Neighbors uses 5 neighbors and the standard Euclidean distance to measure similarity. The Support Vector Machine applies the RBF kernel with a default penalty and scaling for the gamma parameter. Random Forest is configured with 100 decision trees using the Gini criterion to measure splits. Similarly, Extreme Gradient Boosting and Gradient Boosting

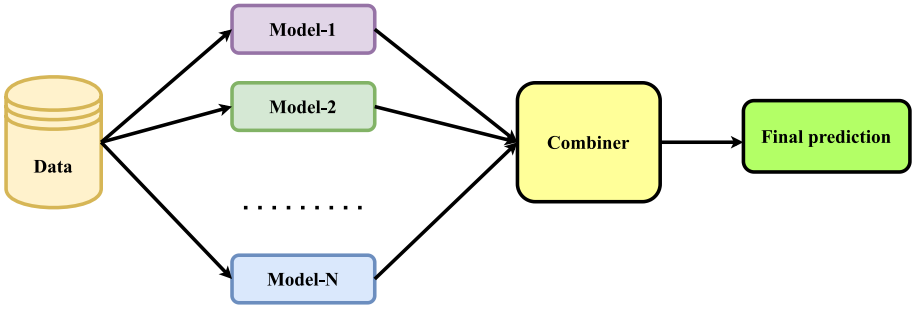


Fig. 7 Voting Ensemble Procedure

Table 5 Models and their hyperparameter

Model	Hyperparameters
Logistic Regression (LR)	Penalty=l2, solver=lbfgs, C=1.0, max_iter=100
K-Nearest Neighbors (KNN)	n_neighbors=5, weights=uniform, metric=minkowski (p=2)
Support vector machine (SVM)	C=1.0, kernel=rbf, gamma=scale
Random forest (RF)	n_estimators=100, criterion=gini, max_depth=None, min_samples_split=2
Extreme gradient boosting (XGB)	n_estimators=100, learning_rate=0.1, max_depth=3, subsample=1, colsample_bytree=1
Gradient boosting (GB)	n_estimators=100, learning_rate=0.1, max_depth=3
AdaBoost (ADB)	n_estimators=50, learning_rate=1.0, base_estimator=DecisionTreeClassifier(max_depth=1)
Gaussian Naïve Bayes (GNB)	var_smoothing=1e-9

both have 100 estimators, a learning rate of 0.1, and a maximum tree depth of 3, while Extreme Gradient Boosting also specifies subsampling and column sampling. AdaBoost uses 50 estimators and builds simple decision trees of depth 1 as base learners. Lastly, Gaussian Naïve Bayes is set with a very small smoothing value to handle numerical stability.

In short, this table gives a clear snapshot of the learning settings used for each model to ensure fair and consistent evaluation.

4 Result analysis

The performance of the suggested model is examined and contrasted with a number of cutting-edge methods that have already been used on the Cleveland Heart Disease dataset in this section.

4.1 Performance metrics

A wide range of performance measures were used in this study to fully assess the ML models. These included classification metrics such as accuracy, precision, recall, F1-score, MCC, ROC-AUC, and Kappa., as well as regression metrics like MSE, MAE, and RMSE. A comprehensive investigation of model performance from several angles is made possible by the use of this wide variety of evaluation criteria. Table 6 provides detailed descriptions of each metric, while Table 7 outlines the corresponding formulas used for their calculation.

Experiments using different machine learning algorithms to predict the existence of cardiovascular disease produced the results shown in this section. In this study, multiple experiments were carried out: The models were trained with different data balancing techniques such as SMOTE, SMOTETomek, and SMOTE+RUS. The performance of each experiment was calculated.

4.2 Model performance analysis

Table 8 provides a comprehensive comparison of model performances under various data balancing strategies within the framework of our proposed methodology. The evaluation includes both traditional ML algorithms such as LR, KNN, SVM, and GNB and more advanced ensemble methods like RF, XGB, GB, and ADB. In addition, two

Table 6 Performance Metrics with Descriptions

Metric	Description
Accuracy (Acc)	Gives a general evaluation of the model's performance by calculating the percentage of correctly identified instances relative to all occurrences
Precision (P)	The model's ability to prevent false positives is demonstrated by the percentage of projected positive cases that are actually positive
Recall (R)	Indicate the percentage of real positive cases that the model accurately detects
F1-Score	It balances the trade-off between recall and precision by taking the harmonic mean of both
Kappa (κ)	Measures agreement between predictions and ground truth, considering the chance agreement
MSE	calculates the average of the squared deviations between the actual and expected values, highlighting more significant errors and frequently utilized in regression projects
MAE	evaluates the accuracy of regression models by calculating the average absolute difference between the actual and predicted values
RMSE	Square root of MSE, giving an error measure in the same unit as the data
ROC-AUC	Better performance is indicated by a greater Area Under the ROC Curve (AUC), which gauges a classifier's capacity to discriminate between classes across a range of threshold values
MCC	Measures overall classification performance, even for imbalanced datasets

Table 7 Equations for performance Metrics

$$\text{Acc} = \frac{TP+TN}{TP+TN+FP+FN} \quad (5)$$

$$P = \frac{TP}{TP+FP} \quad (6)$$

$$R = \frac{TP}{TP+FN} \quad (7)$$

$$F1 = 2 \times \frac{P \times R}{P+R} \quad (8)$$

$$\kappa = \frac{p_o - p_e}{1-p_e} \quad (9)$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \quad (10)$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \quad (11)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \quad (12)$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} (13)$$

ensemble strategies were tested: ENAM, which combines all base models, and ENB3, which uses only the top three best-performing models.

To evaluate performance thoroughly, each model was tested using four different data settings: the original imbalanced dataset, and three popular resampling techniques SMOTE, SMOTETomek, and SMOTE combined with random undersampling (SMOTE + RUS). The model was evaluated using a comprehensive set of evaluation metrics.

4.2.1 Results on the imbalanced dataset

When trained on the original imbalanced dataset, Random Forest (RF) performed the best among all individual models. It achieved the highest accuracy (90.41%), F1-score (90.14), and MCC (81.66), showing that it could handle class imbalance fairly well. Its AUC score of 96.78 also indicated strong classification power. Ensemble models like ENAM and ENB3 performed similarly in terms of accuracy but had slightly lower F1-scores and MCCs. Logistic Regression, XGBoost, and SVM also performed reasonably well, with accuracies around 89% and strong AUC scores. KNN showed the lowest performance in this scenario, with 86.30% accuracy and an MCC of 72.31, indicating that it struggled more with class imbalance.

4.2.2 Results after applying SMOTE

The SMOTE technique improved the performance of most models by creating synthetic samples for the minority class. Random Forest continued to perform well, and

Table 8 Model Performance Summary

Data Balance	Model	Acc	P	R	F1	Kappa	MSE	MAE	RMSE	AUC-ROC	MCC
NONE	LR	89.04	90.32	84.85	87.5	77.76	0.11	0.11	0.33	94.02	77.88
	KNN	86.3	87.1	81.82	84.38	72.2	0.14	0.14	0.37	93.37	72.31
	SVM	87.67	90	81.82	85.71	74.91	0.12	0.12	0.35	96.29	75.18
	RF	90.41	84.21	96.97	90.14	80.9	0.1	0.1	0.31	96.78	81.66
	XGB	89.04	90.32	84.85	87.5	77.76	0.11	0.11	0.33	97.35	77.88
	GB	87.67	87.5	84.85	86.15	75.05	0.12	0.12	0.35	96.29	75.08
	ADB	87.67	85.29	87.88	86.57	75.18	0.12	0.12	0.35	96.02	75.21
	GNB	90.41	88.24	90.91	89.55	80.7	0.1	0.1	0.31	94.32	80.73
	ENAM	90.41	88.24	90.91	89.55	80.7	0.1	0.1	0.31	96.82	80.73
	ENB3	**90.41**	**88.24**	**90.91**	**89.55**	**80.7**	**0.1**	**0.1**	**0.31**	**97.2**	**80.73**
SMOTE	LR	90.41	90.62	87.88	89.23	80.59	0.1	0.1	0.31	94.32	80.62
	KNN	86.3	81.08	90.91	85.71	72.64	0.14	0.14	0.37	92.80	73.08
	SVM	90.41	93.33	84.85	88.89	80.49	0.1	0.1	0.31	97.27	80.77
	RF	91.78	86.49	96.97	91.43	83.58	0.08	0.08	0.29	95.57	84.09
	XGB	89.04	90.32	84.85	87.5	77.76	0.11	0.11	0.33	96.29	77.88
	GB	89.04	85.71	90.91	88.24	78.00	0.11	0.11	0.33	96.21	78.11
	ADB	90.41	86.11	93.94	89.86	80.80	0.1	0.1	0.31	97.08	81.07
	GNB	89.04	85.71	90.91	88.24	78.00	0.11	0.11	0.33	94.47	78.11
	ENAM	90.41	86.11	93.94	89.86	80.80	0.1	0.1	0.31	97.27	81.07
	ENB3	**98.63**	**100**	**96.97**	**98.46**	**97.23**	**0.01**	**0.01**	**0.12**	**99.13**	**97.26**
SMOTETomek	LR	90.41	90.62	87.88	89.23	80.59	0.1	0.1	0.31	94.55	80.62
	KNN	86.3	81.08	90.91	85.71	72.64	0.14	0.14	0.37	91.70	73.08
	SVM	89.04	93.1	81.82	87.1	77.64	0.11	0.11	0.33	97.35	78.13
	RF	90.41	84.21	96.97	90.14	80.90	0.1	0.1	0.31	96.55	81.66
	XGB	91.78	88.57	93.94	91.18	83.50	0.08	0.08	0.29	98.03	83.62
	GB	90.41	86.11	93.94	89.86	80.80	0.1	0.1	0.31	98.41	81.07
	ADB	89.04	85.71	90.91	88.24	78.00	0.11	0.11	0.33	96.63	78.11
	GNB	89.04	85.71	90.91	88.24	78.00	0.11	0.11	0.33	94.70	78.11
	ENAM	90.41	86.11	93.94	89.86	80.80	0.1	0.1	0.31	97.73	81.07
	ENB3	**94.52**	**93.94**	**93.94**	**93.94**	**88.94**	**0.05**	**0.05**	**0.23**	**98.89**	**88.94**
SMOTE + RUS	LR	89.04	87.88	87.88	87.88	77.88	0.11	0.11	0.33	94.02	77.88
	KNN	83.56	80	84.85	82.35	66.99	0.16	0.16	0.41	92.61	67.09
	SVM	86.3	87.1	81.82	84.38	72.20	0.14	0.14	0.37	96.52	72.31
	RF	91.78	90.91	90.91	90.91	83.41	0.08	0.08	0.29	96.29	83.41
	XGB	90.41	86.11	93.94	89.86	80.80	0.1	0.1	0.31	98.11	81.07
	GB	91.78	88.57	93.94	91.18	83.50	0.08	0.08	0.29	97.35	83.62
	ADB	91.78	88.57	93.94	91.18	83.50	0.08	0.08	0.29	97.08	83.62
	GNB	89.04	85.71	90.91	88.24	78.00	0.11	0.11	0.33	94.24	78.11
	ENAM	91.78	88.57	93.94	91.18	83.50	0.08	0.08	0.29	97.27	83.62
	ENB3	**93.15**	**91.18**	**93.94**	**92.54**	**86.21**	**0.07**	**0.07**	**0.26**	**96.67**	**86.24**

ensemble methods saw even greater improvement. ENB3 stood out by achieving near-perfect scores: 98.63% accuracy, 98.46 F1-score, 100% precision, and an MCC of 97.26. This result highlights the strength of ensemble learning when combined with balanced data. Other models like XGB, GB, and ADB also saw performance boosts, achieving high scores across all metrics. KNN, while improved slightly, still remained behind the other models.

4.2.3 Results with SMOTETomek

Using SMOTETomek, which balances the data and removes overlapping or noisy samples, resulted in very consistent performance across many models. RF, GB, ADB, and

XGB all achieved similar high results, with 91.78% accuracy and strong F1-scores and MCC values. XGBoost slightly edged out the others in AUC and MCC. Once again, ENB3 delivered the best overall performance, achieving 94.52% accuracy, 93.94% F1-score, and an MCC of 88.94. KNN, however, continued to lag behind, showing limited benefit from this balancing method.

4.2.4 Results with SMOTE + random undersampling

In the final setup using SMOTE + RUS, there was a slight drop in overall model performance compared to SMOTE alone or SMOTETomek. Despite that, ENB3 still achieved top scores with 93.15% accuracy and an MCC of 86.24. Random Forest also continued to perform reliably, along with XGB, GB, and ADB, all delivering high and balanced performance. SVM and Logistic Regression showed moderate results. KNN once again had the lowest accuracy (83.56%) and MCC (67.09), confirming its relative weakness in dealing with class imbalance even when balanced data is used.

4.2.5 Summary and insights

Overall, these results clearly demonstrate that data balancing significantly improves model performance. SMOTE and SMOTETomek, in particular, were very effective in helping the models make more accurate and balanced predictions. Among all models tested, the ENB3 ensemble consistently outperformed the rest across every evaluation scenario. It was especially impressive under the SMOTE-balanced data, where it reached nearly perfect results. Random Forest and XGBoost also proved to be strong, consistent performers on their own. These findings emphasize that combining the right machine learning models with effective data preprocessing strategies, like resampling, can greatly improve prediction accuracy, especially in sensitive fields like healthcare, where imbalance is a common challenge.

4.3 ROC-AUC curve analysis

In figure 8, the ROC-AUC curve illustrates a clear view of how well each model distinguishes between the two classes. Almost all the models perform really well, staying far above the diagonal "chance" line, which means they are making accurate predictions most of the time. The ensemble of the best three models stands out with an impressive AUC of 0.9913, showing it can almost perfectly separate positive and negative cases. Models like the Support Vector Machine, AdaBoost, and Random Forest also perform exceptionally well, while even the lowest performer, K-Nearest Neighbors, still achieves a solid AUC of 0.9280. Overall, the curves cluster near the top-left corner, highlighting that these models have high true positive rates with very few false positives, and the ensemble approach gives a clear boost in prediction accuracy compared to individual models.

4.4 4-Fold cross validation

The model's performance was evaluated using 4-fold cross-validation, where in each iteration three folds were used for training and the remaining fold for testing. Table 9 summarizes the results, reporting accuracy, precision, recall, F1-score, and AUC-ROC for each fold. Accuracy ranged from 93.15% to 98.63%, indicating consistently strong overall performance. Precision reached 100% in folds 3 and 4, demonstrating the model's

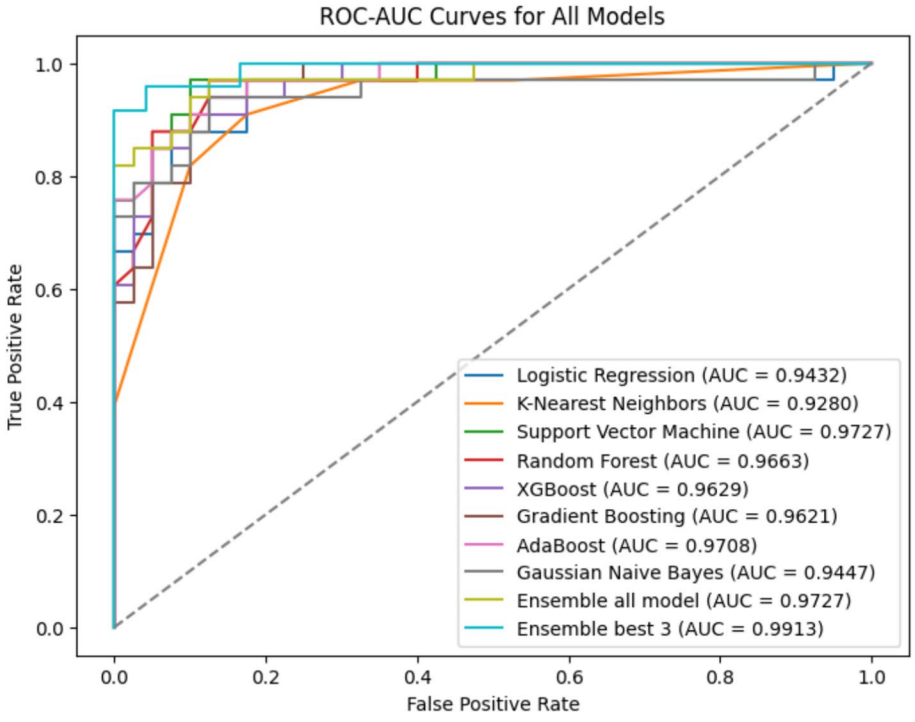


Fig. 8 ROC-AUC curve of different models

Table 9 4-Fold cross validation performances

Fold	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Fold 1	93.15	88.89	96.97	92.75	97.50
Fold 3	98.63	100.00	96.97	98.46	99.13
Fold 4	97.26	100.00	93.94	96.88	97.08
Fold 5	95.89	96.88	93.94	95.38	95.98
Average	96.23	96.44	95.46	95.87	97.42

Table 10 Accuracy Comparison of each balancing techniques

Data Balance Techniques	Accuracy (%)
NONE	90.41
SMOTE	**98.63**
SMOTETomek	94.52
SMOTE + RUS	93.15

ability to correctly identify positive cases, while recall varied between 93.94% and 96.97%, reflecting reliable sensitivity to actual positive instances. The F1-score remained high across all folds, averaging 95.87%, showing balanced performance between precision and recall. AUC-ROC values ranged from 95.98% to 99.13%, with an average of 97.42%, highlighting excellent discriminative capability. Overall, these results indicate that the model performs robustly and consistently across all folds.

4.5 Accuracy comparison

Table 10 and Fig. 9 illustrates the impact of various data balancing techniques on the accuracy of cardiovascular disease prediction. When no balancing method was used, the model achieved an accuracy of 90.41%, indicating potential bias due to the dominance

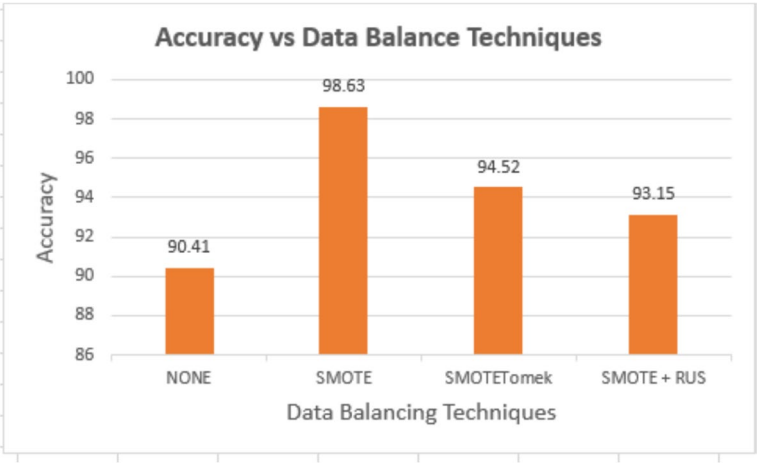


Fig. 9 Accuracy comparison of different data balance techniques

of the majority class. Upon applying SMOTE (Synthetic Minority Oversampling Technique), the accuracy dramatically increased to 98.63%, showcasing its strong effectiveness in generating synthetic samples for the minority class and allowing the model to learn more balanced patterns. SMOTETomek, which combines SMOTE with Tomek links to clean overlapping samples, also improved performance with an accuracy of 94.52%. Meanwhile, the combination of SMOTE with Random Undersampling (RUS) achieved 93.15% accuracy, showing a moderate improvement but slightly less effective than SMOTETomek and SMOTE alone. These results highlight the importance of addressing data imbalance in medical datasets, with SMOTE standing out as the most impactful technique for improving model accuracy in this study.

4.6 Confusion matrix analysis

The impact of different data balancing techniques on the model's classification performance is visually represented through the confusion matrices shown in Fig. 10. Each matrix highlights how well the model differentiates between the "Disease" and "No Disease" classes in terms of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). The four configurations (a) without any data balancing, (b) with SMOTE, (c) with SMOTETomek, and (d) with SMOTE+RUS - demonstrate the progressive improvement in classification as the dataset becomes more balanced.

(a) Without Any Balancing:
In the imbalanced setting, the model produced 36 true negatives and 30 true positives, alongside 4 false positives and 3 false negatives. While the results are generally strong, the slightly higher number of misclassifications (particularly false positives) indicates that the model was somewhat biased toward the majority class. This reflects a common issue when training with imbalanced datasets: under representing the minority class (disease) can lead to subtle but important misclassifications.

(b) With SMOTE: After applying SMOTE (Synthetic Minority Oversampling Technique), the classifier showed notable improvement. The false positives dropped to 0, and false negatives were reduced to just 1. The model correctly identified all 40 non-disease cases and 32 out of 33 disease cases. This excellent balance demonstrates SMOTE's

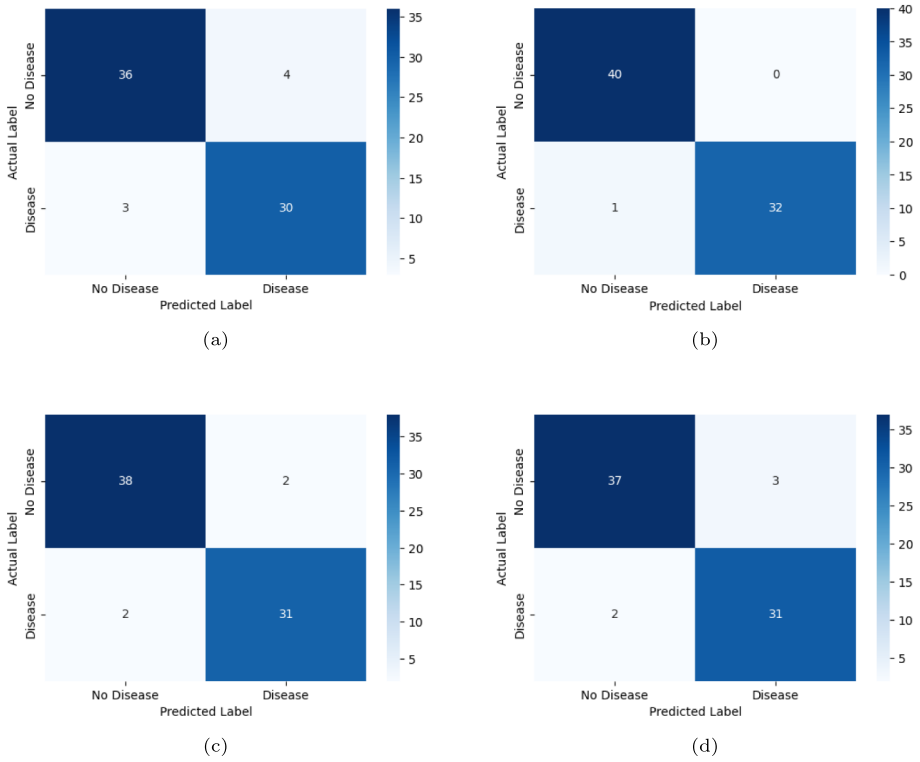


Fig. 10 Confusion Matrix

effectiveness in synthetically generating realistic minority class examples that help the model generalize better, especially in identifying rare but critical disease cases.

(c) With SMOTETomek: Using the SMOTETomek approach, which balances the dataset while also cleaning borderline examples, the confusion matrix shows 38 TN, 31 TP, 2 FP, and 2 FN. This method offered a strong balance between precision and recall. Although slightly more conservative than SMOTE alone, it still significantly reduced misclassification errors compared to the original imbalanced data, indicating its usefulness in refining class boundaries.

(d) With SMOTE + Random Undersampling (RUS): In this combined approach, the model achieved 37 TN and 31 TP, with just 3 FP and 2 FN. While the performance remains solid, it slightly trails SMOTE in precision. The higher number of false positives compared to (b) and (c) may stem from the random undersampling process removing some informative majority class samples. Nevertheless, the classifier maintains reliable overall performance, and this method serves as a viable strategy when computational cost or data volume is a concern.

Summary: Overall, the progression from the imbalanced dataset to balanced versions clearly illustrates how data balancing improves the model's diagnostic accuracy. Among all methods, SMOTE (Fig. 10b) resulted in the best confusion matrix, minimizing both false positives and false negatives. This finding further reinforces the importance of carefully handling class imbalance, especially in critical applications like cardiovascular disease prediction.

4.7 Model explanation

In healthcare, the reliability and transparency of predictive models are essential. While machine learning models can achieve remarkable accuracy in diagnosing diseases or predicting outcomes, their "black-box" nature often makes them difficult to interpret. This lack of explainability can be a major barrier to clinical adoption, as healthcare professionals need to understand the reasoning behind a model's predictions before they can confidently rely on them in decision-making. To bridge this gap, Explainable AI (XAI) techniques are increasingly being used to shed light on how models make decisions. In this study, we utilized two popular XAI methods LIME and SHAP to enhance the interpretability of our cardiovascular disease prediction model and provide meaningful insights into individual and overall feature contributions.

4.7.1 Explanation with LIME

Through the development of straightforward, interpretable models that roughly mimic the behavior of intricate "black-box" models in the vicinity of a particular instance, LIME is an approach that explains individual predictions of machine learning models. It is particularly helpful for interpreting inscrutable models, such as ensemble methods, deep learning models, or any other model that lacks inherent interpretability [36].

Figure 11 provides detailed explanations of the proposed model's predictions: (a) for class 0 (no CVD) and (b) for class 1 (CVD present). The two LIME plots allow us to understand individual predictions for cardiovascular disease (CVD) classification. The first plot shows a prediction for class 0 (no CVD). Here, the model's decision is largely influenced by attributes such as `ca<= 0.00` (no big blood vessels involved), `cp<= 0.67` (less severe chest pain), `slope = 0.00` (flat ST segment), and `oldpeak<= 0.13` (low ST depression). All these are negative indicators of CVD and result in a healthy prediction. Although features like `thal = 1.00` and `sex = 1.00` have some positive effect on CVD, they are not strong enough to alter the final prediction. `Age`, `exang`, and `fbs` make small contributions with minimal effect.

The second graph is a prediction of class 1 (CVD present). The model shows that `ca> 0.33` (more vessels affected), `cp> 0.67` (typical angina), `slope> 0.50` (upsloping ST segment), `sex = 1.00` (male), and `exang = 1.00` (exercise-induced angina) are strong signs of CVD. These features help the model make a lot of decisions. The only important negative factor is `thal<= 0.00`, which slightly moves the prediction away from CVD, but not enough to change the result. Other features such as `oldpeak`, `fbs`, and `age` have less impact.

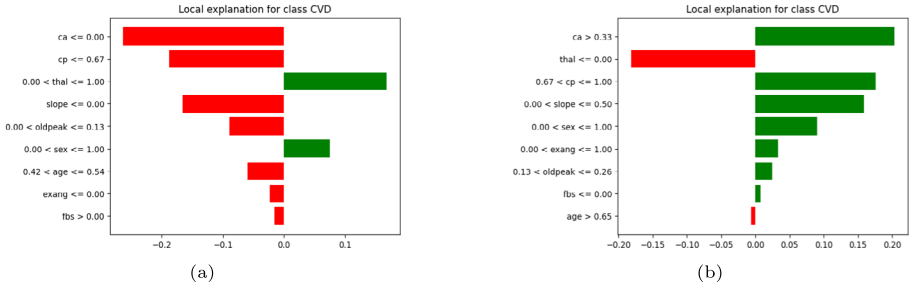


Fig. 11 LIME Explanation (**a**) with class 0 and (**b**) with class 1

4.7.2 Explanation with SHAP

An XAI technique called SHAP seeks to provide justifications for the predictions generated by ML models. SHAP theory, based on game theory, gives each feature a value representing its contribution to a particular prediction. Positive SHAP values mean the feature pushes the prediction higher, meaning toward class 1, and negative SHAP values mean it pushes the prediction lower, thereby toward class 0. Basically speaking, SHAP is very useful for both individual prediction analysis (local explanation) and model understanding in general (global explanation), especially in cases of application as high-stakes as healthcare [37].

Figure 12 demonstrates the SHAP summary plots used to explain individual predictions made by the proposed model. Subfigure (a) refers to a prediction for class 0 (non-CVD), while subfigure (b) shows a prediction for class 1 (CVD). The low values of the main features such as *ca*, *cp*, *thal*, *slope*, and *oldpeak*, which have all negative SHAP values, are primarily influencing the prediction in subfigure (a) toward class 0. Some high values (for example, in *thal* or *cp*) would have slightly favored the prediction to become class 1, but their influence is too weak for any change. Conversely, subfigure (b) shows a prediction in which the model senses the presence of CVD (class 1). In this case, strong high values of *ca*, *cp*, *thal*, and *slope* have positively contributed with strong positive SHAP values for the CVD prediction. It is also confirmed through other features, such as *sex* and *exang*. The overall picture from the SHAP plots clearly reveals how the feature values push the model either towards non-CVD or CVD prediction.

4.8 Comparison with related works

The proposed research, provides an all-around framework to increase prediction efficiency and interpretability for cardiovascular disease diagnosis. Paying attention to prominent factors such as optimal feature selection, managing class imbalance, robust ensemble modeling, and explainable AI procedures, the model aims to deliver a valid and clear decision-support system for health application.

Table 11 depicts a comparative analysis of the proposed cardiovascular disease prediction model with some of the latest state-of-the-art studies. Korial et al. [13] utilized Chi-square for feature selection and a voting ensemble model with 92.11% accuracy without data balancing and explainability. Kumar et al. [14] and Chandrasekhar et al. also utilized ensemble-based methods but without feature selection, data balancing, and XAI and achieved accuracies of 92.30% and 93.44%, respectively. Abdellatif et al. [16] used Extra Trees and SMOTE to get 95.73% accuracy, while Rahim et al. [18] used SMOTE and feature importance to get 95.50%, with both having no XAI integration. Tama et

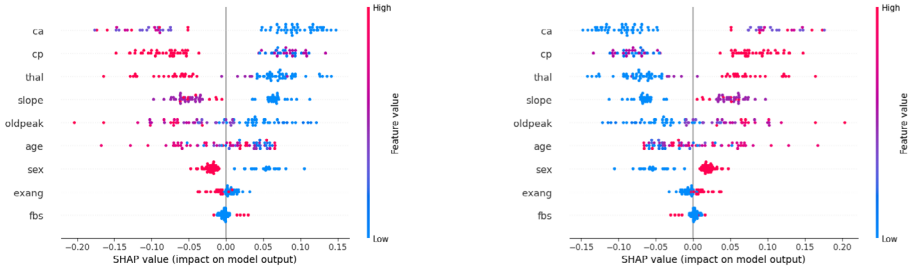


Fig. 12 SHAP Explanation (**a**) with class 0 and (*b*) with class 1

Table 11 Comparison of proposed work with some other recent work in terms of accuracy

Author	Feature Selection	Number of Features	Data Balancing	Model	Accuracy	XAI
korial et al. [13]	Chi-square	5	X	Voting ensemble ML	92.11%	X
kumar et al. [14]	X	X	X	Stacked ensemble model	92.30%	X
Chandrasekhar et al. [15]	X	X	X	Voting ensemble ML	93.44%	X
Abdellatif et al. [16]	X	X	SMOTE	Extra Trees	95.73%	X
Tama et al. [17]	PSO	7	X	Stacked ensemble model	85.86%	X
Rahim et al. [18]	Feature importance	6	SMOTE	Ensemble	95.50%	X
Noroozi et al. [19]	filter, wrapper, and evolutionary	7	X	SVM	85.50%	X
Ahamad et al. [20]	X	X	X	SVM	87.91%	X
Ogundepo et al. [21]	Chi-square	8	X	SVM	85.00%	X
Shrestha et al. [22]	X	X	X	XGBoost	90.00%	SHAP
Nursyahrina et al. [23]	RNN	9	X	RNN	88.52%	X
Nazari et al. [24]	X	X	X	Ensemble	88.05%	X
Reddy et al. [25]	X	10	X	GNB	86.00%	X
Proposed Model	XGBoost	9	SMOTE, SMOTE-Tomek, SMOTE+RUS	Voting Ensemble	**98.63%**	SHAP LIME

al. [17] used PSO with a stacked ensemble model to get 85.86% accuracy. Noroozi et al. [19] combined filter, wrapper, and evolutionary feature selection and SVM and achieved 85.50%, Ahamad et al. and Ogundepo et al. used SVM with minimal pre-processing and achieved 87.91% and 85.00%, respectively. Shrestha et al. [22] used XGBoost with the inclusion of SHAP to include explainability and achieved 90.00%. Nursyarhina et al. [23] used RNN while Nazari et al. [24] used ensemble models in isolation, without feature selection and XAI, and achieved accuracies of 88.52% and 88.05% respectively. Reddy et al. [25] applied GNB with ten features, getting 86.00%. But the model in this paper applies XGBoost as a feature selection mechanism and extracts 9 best features and uses multiple data balancing technique with SMOTE, SMOTETomek, and SMOTE+RUS. Strong voting-based voting ensemble model is used, and it gives a higher accuracy of 98.63%. Moreover, the model uses both SHAP and LIME for interpretability, giving understandable and transparent results. This combination of cutting-edge feature selection, complete spectrum data balancing, stable ensemble learning, and double XAI integration makes the suggested model extremely accurate, trustworthy, and worthy of clinical decision making.

Abdellatif et al. [16] report very high accuracy (95.73%) using only SMOTE and Extra Trees, without the additional steps in our pipeline. However, our goal was not just to push accuracy slightly higher, but to build a pipeline that is more robust, generalizable, and clinically useful. Each stage plays a role: preprocessing cleans the data, feature selection reduces noise, balancing addresses class imbalance, ensemble learning improves stability, and XAI makes the results transparent for clinicians. While individual methods may achieve high accuracy, combining all five steps ensures that the model is not only

accurate but also reliable and interpretable-qualities that are essential for real-world use in cardiovascular disease prediction.

4.9 Validation of the research questions

We validate the research questions using the following important standards to make sure they are pertinent, well-structured, and in line with the goals of the study:

- RQ1: The XGBoost feature selection strategy prioritized the importance of attribute classification according to its contribution to modeling performance. The application of the 9 most relevant selected features led to a notable increase in prediction accuracy.
- RQ2: Oversampling methods, like SMOTE, Undersampling methods, like Random Under Sampling, or a hybrid technique using SMOTETomek and SMOTE+RUS, were used to improve the class imbalance classifying process. These methods have reduced bias interpretation in the model toward the prevalent class by balancing the database.
- RQ3: SMOTE-RUS is an effectual approach out of those tried, which improves class balance with increasing diversity of data. The individual oversampling and undersampling approaches did not match its accuracy and AUC-ROC.
- RQ4: In terms of accuracy (98.63%) and AUC-ROC (99.85%), the voting ensemble model that combined many classifiers proved to be superior to an individual model in prediction due to its superior generalization and robustness to cardiovascular illness.

The solutions to the research questions explain how feature selection, data balance, and ensemble learning can improve the prediction of cardiovascular disease (CVD). While class imbalance through hybrid methods (SMOTE+RUS) improves prediction accuracy greatly, choosing the most pertinent features improves model efficiency and interpretability. The hybrid method outperformed other data-balancing techniques in reducing bias and increasing classification performance. The ensemble voting model, with its greater accuracy and generalization than individual classifiers, has the potential to be an effective method of estimating CVD risk in a real-world environment. The findings support the study's goals and show how machine learning may be used to better prevent and identify cardiovascular disease early.

5 Conclusion

This research introduces an integrative and interpretable ML framework for the precise prediction of CVD, tackling significant problems like outliers, irrelevant features, class imbalance, and limited interpretability of traditional ML models. The suggested method incorporates data preprocessing, feature selection, data balancing, ensemble learning, and XAI techniques to improve the performance of the model as well as its transparency. Then, Z-score-based outlier removal was applied to remove anomalies and improve data quality. This was followed by feature selection from the XGBoost (XGB) feature importance method, which chose the 9 most significant features, reducing dimension and improving model efficiency. The Class imbalance was tackled using three advanced resampling methods: SMOTE, SMOTETomek, and SMOTE+RUS to ensure the models were able to learn properly from the majority and also minority classes.

This study explored eight well-known supervised learning algorithms namely LR, KNN, SVM, RF, XGB, GB, ADB, and GNB. Each model was evaluated using a wide range of performance metrics. Among all the models tested RF, LR, and ADB consistently delivered the best results. Based on their strong performance, we combined these three into a voting ensemble model, which significantly outperformed the individual models. The ensemble achieved a top accuracy of 98.63% and an outstanding ROC-AUC score of 99.13% on the Cleveland Heart Disease dataset, using just 9 optimized features. These results surpass many existing methods reported in current research, showing the strength and reliability of our approach.

To make the model's decisions more transparent and trustworthy, we integrated XAI techniques, specifically SHAP and LIME. These tools helped highlight which features had the biggest impact on predictions and explained how the model made decisions on a case-by-case basis. In the healthcare industry, where knowing the "why" behind a prognosis can be just as significant as the prediction itself, this extra layer of interpretability is extremely crucial. This interpretability is essential for clinical adoption and risk-based decision-making.

In this study, ensemble methods were selected for cardiovascular disease (CVD) prediction due to their demonstrated ability to achieve superior accuracy and robustness in handling complex and imbalanced datasets, thereby lowering the risk of misclassification. To promote clinical trust and usability, explainable AI (XAI) techniques were integrated, offering transparent insights into the most influential risk factors and making the model's predictions interpretable and actionable for medical decision-making.

In conclusion, the proposed framework offers a powerful, accurate, and transparent solution for early cardiovascular disease prediction. It supports risk stratification and aids in timely medical intervention, ultimately contributing to improved patient outcomes and reduced CVD-related mortality. This work lays a strong foundation for implementing reliable AI-driven tools in real-world healthcare settings and can be further extended to other medical prediction tasks.

6 Limitations and future study

Despite the encouraging results, this study has a few limitations that need to be acknowledged. The model was trained and validated using the Cleveland Heart Disease dataset, which, although widely used in research, contains only 303 patient records and may not fully represent the diversity of patients seen in real-world clinical settings. Differences in demographics, comorbidities, and lifestyle factors across larger, more heterogeneous populations could affect how well the model generalizes. The small sample size also raises the risk of overfitting, even though data balancing and cross-validation strategies were applied.

Moreover, this study mainly relied on static clinical features and did not include real-time or longitudinal health data, which limits its ability to capture changes in patient conditions over time. While SHAP and LIME were helpful in making the model's predictions more interpretable, further refinement of these explainability methods-tailored to clinical workflows-could make the outputs more practical and easier for clinicians to use.

Future work should aim to test and refine the model using larger, more diverse, and multi-center datasets that better reflect real-world populations. Including additional

clinical, behavioral, and genetic data, along with real-time information from wearable sensors, could improve predictive accuracy and allow for continuous monitoring. External validation on independent datasets and eventual deployment in real healthcare environments will be essential for proving robustness. Exploring hybrid and deep learning approaches, as well as using AutoML to automate feature selection and model tuning, may further boost performance. Finally, integrating user-friendly decision-support systems into clinical practice and extending the model for long-term risk prediction would improve its usability and overall value for cardiovascular disease prevention and management.

Abbreviations

CVD	Cardiovascular disease
WHO	World Health Organization
R	Recall
SMOTE	Synthetic minority oversampling
LR	Logistic regression
RF	Random forest
DT	Decision tree
ML	Machine learning
SVM	Support vector machine
LSTM	Long short-term memory
XGB	eXtreme gradient boosting
SHAP	Shapley additive explanations
GNB	Gaussian Naive Bayes
GB	Gradient boosting
AUC	Area under curve
CNN	Convolutional neural network
ENAM	Ensemble with all mode
RNN	Rough neural network
DL	Deep learning
Acc	Accuracy
P	Precision
F1	f1 score
Kappa	Cohen's kappa
MCC	Matthews correlation coefficient
MAE	Mean absolute error
MSE	Mean squared error
XAI	Explainable artificial intelligence
ADB	Adaptive boosting
KNN	k-Nearest neighbors
ANN	Artificial neural network
LIME	Local interpretable model-agnostic explanations
NN	Neural networks
MLP	Multi-layer perceptron
ROC	Receiver operating characteristics
PSO	Particle swarm optimization
ENB3	Ensemble with best3
AI	Artificial intelligence
ANN	Artificial neural network

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Data availability
The Heart Disease Dataset is available on https://archive.ics.uci.edu/dataset/45/ heart+disease.

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The authors declare no Conflict of interest.

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